

Exchange rates as quantitative structure on a hierarchy of maps

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Abstract

Classifying a family of maps under an implementation relation produces a preorder and stops there. We add the missing quantitative layer: an *exchange rate* counting how many copies of one map a second map can supply per copy, and we study the geometry it induces. The main result is a structure theorem — every exchange geometry is a one-dimensional geometry plus a defect bounded by $\log 2$, sharply, because the underlying Hilbert projective metric fibres over a line with fibres of diameter exactly one bit. We record what the rate reads off an arithmetic family, why the comparison it induces is not transitive, and how large that failure is. One theorem here — on the square class of a quadratic form — is arithmetic the signature rate provably cannot see. Every numbered claim is marked *proved* or *computed*; sources are the accompanying findings files.

1 The framework is general, and it adds structure rather than objects

Fix any class of maps $\mathcal{P}$ and any notion of **implementation** — a relation $f \preceq g$ meaning “ g can be wired up to compute f ”. Classification of $\mathcal{P}$ under $\preceq$ is the classical activity: it produces a preorder, and its Hasse diagram is the hierarchy. For polynomial maps over a field the standard choice is *affine implementation*, $f = b \circ g \circ a$ with a, b affine, and the resulting posets of quadratic and cubic maps over $\mathbb{F}_q$ are the objects this project started from.

A preorder answers one question — *can g do f at all?* — and is silent on the quantitative one. The exchange rate supplies the missing structure. Writing $f^{\times k}$ for the k -fold Cartesian power and

$$k_{g \rightarrow f}(r) = \max\{k : f^{\times k} \preceq_{\text{aff}} g^{\times r}\},$$

block-diagonal composition gives $k(r+s) \geq k(r) + k(s)$, so Fekete’s lemma yields

$$C_{\text{aff}}(g \rightarrow f) = \lim_{r \rightarrow \infty} \frac{k_{g \rightarrow f}(r)}{r}. \tag{1}$$

This is the object. It refines the hierarchy without changing it: $f \preceq g$ already implies $C_{\text{aff}}(g \rightarrow f) \geq 1$, and C_{aff} grades every comparable pair by *how many* copies, and every incomparable pair by an exchange ratio the preorder cannot see. The construction is generic — it needs only a monoidal product and an implementation relation — so it applies verbatim to any hierarchy of maps a reader already cares about.

The computable proxy is the **signature rate**. The signature $\sigma(f)$ is the multiset of fiber cardinalities; for Cartesian powers fiber sizes multiply, and the entropy method gives the closed form

$$C(g \rightarrow f) = \inf_{0 \leq \beta \leq \infty} \frac{\log Z_g(\beta)}{\log Z_f(\beta)}, \quad Z_a(\beta) = \sum_i a_i^\beta. \quad (2)$$

Here β is an inverse temperature and Z a partition function with energy levels $-\log n_i$; (2) is a theorem about the operational limit, not a second definition. Throughout, C means (2) and C_{aff} means (1).

1.1 The two rates are incomparable

An earlier draft asserted that affine implementation forces signature implementation, hence $C_{\text{aff}} \leq C$. **That is false**, and the counterexample is a one-liner [affine_rate]. Over $\mathbb{F}_3$ take $g = x^2 + y$ and $f = x^2$. Then

$$f = g \circ a \quad \text{with the singular affine processor } a(x, y) = (x, 0),$$

so $f \preceq_{\text{aff}} g$ in one shot and $C_{\text{aff}}(g \rightarrow f) = 1$; while $\sigma(g) = (3, 3, 3)$ and $\sigma(f) = (6, 3)$ give

$$C(\sigma(g) \rightarrow \sigma(f)) = \frac{\log 3}{\log 6} = 0.6131471927654584\dots < 1 = C_{\text{aff}}(g \rightarrow f). \quad (3)$$

The mechanism. The affine model explicitly permits singular input processors. Here a is 3-to-1: it collapses the nine-point source onto a three-point line, and pulling back through it *manufactures* a fiber of size 6 from a map all of whose fibers have size 3. Signature implementation cannot do that — its processors inject each target fiber into an assigned source fiber, so fiber sizes are inherited and never enlarged. Neither rate bounds the other: on the five non-constant affine classes of quadratic maps $\mathbb{F}_3^2 \rightarrow \mathbb{F}_3$, nine ordered pairs have $C_{\text{aff}} < C$ and four have $C_{\text{aff}} > C$.

Worse, C_{aff} is not a function of the signature at all. The maps x and $x^2 + y$ share the signature $(3, 3, 3)$, so their signature rates against x^2 agree to all 40 digits; but $C_{\text{aff}}(x \rightarrow x^2) = 0$ (the x -atoms are affine functions and span nothing of degree 2) and $C_{\text{aff}}(x^2 + y \rightarrow x^2) = 1$. Two pairs, identical C , affine rates at the two ends of the range. *No functor between the two processor categories exists in either direction.*

That the signature loses information was already visible on the arithmetic side: over $\mathbb{F}_{11}$, 1744 genus-two pencils $y^2 = P(x) + c$ fall onto only 296 signatures, and every one of the 209 multiply-realised signatures comes from pencils with different fiberwise isogeny data (419 of 420 at $\mathbb{F}_{13}$) [flux_arithmetic]. But the affine counterexample is stronger than lossiness: C is not an approximation to C_{aff} from either side. **Every statement below is a statement about C , and the word “signature” belongs in each of them.**

This looks like a liability, and §2.1 shows it is the opposite: the difference between the two rates is exactly where the arithmetic that C cannot see is kept.

2 What the rate reads off an arithmetic family

Let $f : \mathbb{A}^2 \rightarrow \mathbb{A}^1$ over $\mathbb{F}_q$ have geometrically irreducible fibers, $N_c = \#f^{-1}(c)$, $a_c = q - N_c$ the Frobenius traces, $m_2 = q^{-2} \sum_c a_c^2$, and let $L = (q, \dots, q)$ be the flat signature of a linear map [m_and_e_and_a_c].

Proposition 2.1 (proved). $C(L \rightarrow f) = \log q / \log \max_c N_c$, attained at $\beta = \infty$.

Proposition 2.2 (proved). $1 - C(f \rightarrow L) = (3 - 2\sqrt{2}) m_2 / (2q \log q) + O(q^{-3/2})$, with the infimum at

$$\beta^* = \sqrt{2} - 1 = 0.414213562 \dots$$

independent of family, genus and q .

So the two ends of the temperature axis read two classical statistics: the **extreme Frobenius trace** at $\beta = \infty$ and the **second moment** at β^* . The interior carries the whole moment ladder, each order damped by $\approx 0.6/\sqrt{q}$. Flatness is equivalent to a permutation-polynomial condition, which subsumes the known $q \equiv 2 \pmod{3}$ result and reaches non-congruence conditions such as supersingular primes.

Two things are provably invisible, for structural reasons rather than by accident: the *smallest* fiber, because isolating it needs $\beta < 0$ where Z is no longer order-preserving; and the Katz–Sarnak symmetry type, because $\sum_c a_c = 0$ identically and that first moment is exactly the separating statistic.

Remark 2.3 (the objection, stated plainly). $C(L \rightarrow f)$ is a strictly monotone function of $\max_c N_c$, and $1 - C(f \rightarrow L)$ is to leading order affine in m_2 . The rate is a function of the signature and the signature is the list $\{N_c\}$, so nothing the rate reports about arithmetic could fail to be computable from $\{N_c\}$ directly. Information flows into the matrix, not out of it. What survives is not a new theorem about curves but a statement about the *comparison*: of all ways to compress $\{N_c\}$, resource-theoretic comparison against a linear map returns exactly the extreme value and the second moment, and returns them at two specific temperatures. That is a consistency check on the framework, and it is how it should be written.

2.1 What the affine rate reads, and the signature rate cannot

The objection above is fatal to C and does not touch C_{aff} . The affine rate is not a function of the signature (§1), so nothing forces its arithmetic content to be recoverable from $\{N_c\}$ — and it is not.

Theorem 2.4 (proved). [affine_rate/ISOTROPY] *Let Q be a binary quadratic form over any field K with $\text{char } K \neq 2$. Then*

$$N_k(Q \rightarrow x) = \lceil \frac{3k}{2} \rceil, \quad C_{\text{aff}}(Q \rightarrow x) = \frac{2}{3} \quad \text{if } Q \text{ is anisotropic;}$$

$$N_k(Q \rightarrow x) = k, \quad C_{\text{aff}}(Q \rightarrow x) = 1 \quad \text{if } Q \text{ is isotropic.}$$

Nothing in the proof is finite-field-specific: the only arithmetic fact used is that anisotropy forces an atom’s linear part to stay inside the support of its quadratic part. The statement therefore holds over $\mathbb{Q}_p$ and over $\mathbb{R}$ as well — new for the seven anisotropic square classes of the 2-adic quadratic-map poset — and it puts the value $2/3$ on a Hasse arrow the finite-field posets record as missing.

Corollary 2.5 (proved). *Over $\mathbb{F}_q$, $C_{\text{aff}}(x^2 + y^2 \rightarrow x) = 2/3$ when $q \equiv 3 \pmod{4}$ and $= 1$ when $q \equiv 1 \pmod{4}$. The uniformly anisotropic family $x^2 - ny^2$ (n a non-square) has $C_{\text{aff}}(\cdot \rightarrow x) = 2/3$ at every odd q .*

So C_{aff} jumps by exactly $1/3$ across a congruence: it detects the **square class of the discriminant**, a genuine arithmetic invariant. What C does on the same pair is the point.

	$q = 3$	$q = 47$	behaviour
C_{aff} separation of the two classes	1/3	1/3	constant
C separation of the same two classes	0.0087	0.0014	$\rightarrow 0$
amplification C_{aff}/C	38	245	grows in q

C moves in the right direction — an isotropic form is the easier resource — but its separation *shrinks* as q grows, while C_{aff} 's is exactly 1/3 for every q . The signature sees (4, 4, 1) against (5, 2, 2) and barely flinches; the affine rate sees a hyperbolic plane against an anisotropic form and jumps.

Remark 2.6 (what this does and does not answer). This is the one place in the programme where the exchange rate carries arithmetic that is not a monotone relabelling of a classical statistic of the same object, and the reason is structural: C_{aff} is a functional of the *maps*, and the signature is a lossy quotient of them. It does not rescue the propositions above, which remain translations of $\max_c N_c$ and m_2 . It relocates the claim: the arithmetic content of this framework lives in C_{aff} , and every computation in this project other than Theorem 2.4 has been performed on C .

3 Main result: one dimension plus a bounded defect

This is the most important thing the programme has produced, because it is the only statement about the *framework* rather than about particular resources.

Write $F_a = \log Z_a$, $u_a = \log F_a$, $R = \log r$ (r = number of fibers), $\Lambda = \log(\text{max fiber})$, $\sigma = \log(R/\Lambda)$, and $s = \log \beta$.

Theorem 3.1 (proved; verified to $3.6 \cdot 10^{-15}$). [realizability]

$$u_a(e^s) = \log \Lambda_a + \max(\sigma_a, s) + w_a(s), \quad 0 \leq w_a \leq \log(1 + e^{-|s - \sigma_a|}),$$

with w_a unimodal, peaked exactly at $s = \sigma_a$, and 1-Lipschitz, of height $\leq \log 2$ with equality iff a is flat. Every profile is a kink plus one bump.

Splitting $-\log C$ into symmetric and antisymmetric parts — $S = d/2$ the exchange metric, A the comparison, with $a \prec b \iff A(a, b) > 0$ — Theorem 3.1 gives

$$d(a, b) = |\sigma_a - \sigma_b| + \varepsilon, \quad 0 \leq \varepsilon \leq \log(1 + e^{-|\Delta\sigma|}) \leq \log 2, \quad (4)$$

$$A(a, b) = \psi(b) - \psi(a) + D, \quad |D| \leq \frac{1}{2} \log(1 + e^{-|\Delta\sigma|}) \leq \frac{\log 2}{2}, \quad (5)$$

with $\psi = \frac{1}{2} \log(\log r \cdot \log M)$. Both constants are **sharp** and both bounds are **independent of the number of points**.

Theorem 3.2 (proved). [realizability/OBSTRUCTION] $d(a, b)$ is exactly the **Hilbert projective metric** between F_a and F_b — of the ambient cone $\mathcal{K}$ of positive functions under the pointwise order, restricted to the achievable set $\mathcal{C} = \{\Phi \text{ convex, nondecreasing, } \Phi \geq \Lambda_\Phi \beta\}$, which is the projective closure of that set. Cartesian powers are the projective rescaling the Hilbert metric quotients out.

The distinction between the two cones matters: d is *not* the intrinsic Hilbert metric of $\mathcal{C}$, and reading it that way yields a contraction theorem that does not exist [birkhoff, §2.5].

Theorem 3.3 (proved). [birkhoff] The constant $\log 2$ is a **projective diameter**. With $\mathcal{C}_0 = \{\Phi \in \mathcal{C} : \Phi(0) = \Lambda_\Phi\}$ the $\sigma = 0$ fibre, $\text{diam}_d \mathcal{C}_0 = \log 2$ exactly, attained between $\max(1, \beta)$ and $1 + \beta$. The β -dilation group acts by isometries with geodesic orbits, σ is a surjective 1-Lipschitz map to $\mathbb{R}$, and every fibre is isometric to $\mathcal{C}_0$.

Equivalently: *the exchange geometry is the real line thickened by exactly one bit*. That is the geometric source of both sharp constants, and the attached Birkhoff ratio is $\tanh(\frac{\log 2}{4}) = 3 - 2\sqrt{2}$ exactly.

3.1 What Theorem 3.2 does not buy

No processor is a Birkhoff contraction (*proved*). Tensoring is a **translation** $\Phi \mapsto \Phi + F_c$, neither linear nor degree-1 homogeneous; and every operation that *is* linear shifts σ by a constant, so its projective diameter is infinite and its Birkhoff ratio is exactly 1, attained. Worse, tensoring **expands** d without bound — $d(a, a^{\otimes k}) = 0$ while $d(a \otimes c, a^{\otimes k} \otimes c) \rightarrow \log k$ — so d is a pseudometric on resources and $\otimes$ does not descend to its quotient. Any claim that conversion loses distinguishability at a definite rate is false. The correct positive statement is that all four resource operations are **nonexpansive in Thompson’s part metric**: Hilbert and Thompson differ exactly along the projective direction, and the projective direction is what conversion trades in.

3.2 Everything previously observed is a corollary

The ℓ_2 -distortion ≈ 1.1 with $c_2/\log n$ *falling* — Bourgain’s $O(\log n)$ is unreachable because the non-line budget does not grow with n . The mild pentagonal failure, and every negative-type violation, confined to a window of diameter $2\log 2$. The measured 4–9% curl of the comparison and its flatness in n from 8 to 698 points — the bounds in (5) are per-edge. The same constant reappearing in the $q \rightarrow \infty$ limit.

Corollary 3.4 (proved). *The geometric mean of the rate asymmetry $C(b \rightarrow a)/C(a \rightarrow b)$ around any preference cycle is at most 2.*

Sharpness is closed on one side. Writing $\text{curl } A = \frac{1}{2} \log \Omega$ with Ω the ratio of forward to backward arbitrage around the triangle (*proved*), the question “is the sharp constant $(\log 2)/2$?” is exactly “*is* $\Omega \leq 2$ — does the forward arbitrage exceed the backward by at most one bit?” The lower bound is proved in closed form by $\Phi_1 = \max(1, \beta)$, $\Phi_2 = 1 + \beta$, $\Phi_3 = 1 + e^T \beta$, giving $\text{curl } A = \frac{1}{2}(\log 2 - \log(1 + e^{-T})) \nearrow (\log 2)/2$ — so the supremum is $\geq (\log 2)/2$ and is *not attained* [birkhoff]. The upper bound remains open between $(\log 2)/2$ and $3(\log 2)/2$, and any proof must use the σ -spread: inside a single fibre the supremum is only $\frac{1}{2} \log(4/3)$.

4 Are the cycles important?

The exchange comparison is not transitive. Over $\mathbb{F}_{11}$ there are 132 strict 3-cycles among genus-two pencils $y^2 = P(x) + c$ in a complete enumeration, 1475 over $\mathbb{F}_{13}$, with certified witnesses over $\mathbb{F}_{11}$ and $\mathbb{F}_{101}$ verified three ways. Substantial effort went into finding and certifying them. **Are they worth it?** Three results weaken the obvious reading, one strengthens it, and one bounds how much can be claimed.

Against: cycles are not arithmetic. A random pool of signatures matched to a curve pool in the only things that matter (q fibers, $\sum_c N_c = q^2$, matched trace distribution) has the *same* curl, and the tightest control has slightly more. Over 108 pools, $\|\text{curl}\|/\|A\|$ is a function of the trace spread to $R^2 = 0.93$ [flux_arithmetic]. Cycles are a property of near-flat signature *shape*, not of the arithmetic that produced it.

Against: cycles mostly do not descend to the maps. The $\mathbb{F}_3$ tensor three-cycles are cycles of *signatures*, and since C_{aff} is not a signature invariant (§1) the question is only well posed after choosing affine orbits. The seven signature cycles lift to twelve orbit triples, and **eight of the twelve are refuted under C_{aff} [affine_rate]** — in each case because one vertex is affinely implementable from another, forcing the opposite comparison on that edge. Where the descent can be tested, it mostly fails.

For, and this replaces an argument we had wrong. An earlier draft claimed cycles are a zero-temperature artefact — that averaging over temperature destroys them. That is false [maslov]. Write $c_\beta(a \rightarrow b) = F_a(\beta)/F_b(\beta)$ for the fixed- β exchange rate and let M_λ be the λ -power mean over a temperature prior ρ . Then the whole comparison is

$$A_\lambda(a, b) = \frac{1}{2} \log \frac{C_\lambda(a \rightarrow b)}{C_\lambda(b \rightarrow a)}, \quad C_\lambda(a \rightarrow b) = M_{-\lambda}(c.(a \rightarrow b)), \quad (6)$$

with $\lambda = \infty$ the framework’s infimum, $\lambda = 1$ the plain arithmetic average of the transition rates, and $\lambda = 0$ the geometric mean. Two facts:

- At a *single* temperature the comparison is the total order of the scalar $F(\beta)$, since $c_\beta(a \rightarrow b)c_\beta(b \rightarrow a) = 1$. No pool cycles at any one temperature; all non-scalar content comes from the β -dependence of the argmin.
- Cycles survive averaging. There are certified 3-cycles of *flat* signatures — which Theorem 3.1 *proves* cannot cycle at $\lambda = \infty$ — living in bounded bands $\lambda \in (1.5211, 2.5376)$ and $(0.8032, 1.3139)$, the second a cycle at $\lambda = 1$ exactly.

The comparison is a potential *iff* $\lambda = 0$: the geometric mean is the unique power mean commuting with reciprocals, so $C_0(a \rightarrow b)C_0(b \rightarrow a) = 1$ and the trade becomes reversible. But the same identity gives $S_\lambda = \lambda \text{Var}_\rho / 2 + O(\lambda^3)$, so at $\lambda = 0$ every distance is zero and every pair freely interconvertible. The scalar complexity available there is the potential of an empty theory.

Remark 4.1. Non-scalarity is not an artefact of the infimum. It is a consequence of **irreversibility**: the comparison fails to be a potential exactly because a power mean of order $\lambda \neq 0$ does not commute with reciprocals. The infimum is the extreme case, not the source.

Where the cycles come from, mechanically. Brandão–Horodecki–Ng–Oppenheim–Wehner govern conversion under thermal operations by a continuum of Rényi α -free energies — the same β -indexed family. Their scalar is an infimum of a **difference**, hence a partial order, acyclic by construction; ours is an infimum of a **ratio**, hence a tournament, which can cycle [quantum]. And the ratio is *forced*: these resources are unnormalised, so the difference diverges like $\beta(\Lambda_a - \Lambda_b)$ and the ratio is the only finite scalar. **Cycles are the price of comparing resources with no common normalisation** — the situation for maps, where nothing plays the role of a fixed particle number.

The hedge, and it is the number to quote. A single *unfitted* temperature prior — uniform in $s = \log \beta$ on $[0, 6]$ — reproduces 99.39% of the tropical order on the $\mathbb{F}_{11}$ arithmetic pool and 99.51% on $\mathbb{F}_{13}$; a fitted prior reaches 99.62% and transfers across q without loss, against an information-theoretic ceiling of 99.90% (at least 42 edge-disjoint 3-cycles must be misordered). Both beat the HodgeRank potential ψ_{opt} at 98.31% [maslov]. So the irreducibly non-scalar part of the comparison is about **four parts in a thousand**.

The defensible claim. Asymptotic convertibility is governed by no scalar complexity, because the comparison is a non-reversible aggregation over temperature. The failure is real but small: a single temperature prior recovers 99.4% of the order, and at $\lambda = \infty$ rates around any preference cycle differ by a geometric-mean factor of at most two. What this does *not* support is the story the cycle hunt was implicitly aimed at — that cycles reveal something arithmetic. They do not.

One genuinely quantum instance. Replacing $\text{Tr } A^\beta$ by a sandwiched Rényi divergence against a background operator gives a non-spectral rate under which three qubit states form a certified 3-cycle ($|\text{curl } A|/\sum |A| = 1$ exactly, 40 digits) whose decohered shadow is transitive [quantum]. Two caveats: the plain quantisation $\text{Tr } A^\beta$ is *empty* — it factors through the spectrum identically, so a quantum cycle without a classical shadow cannot exist — and every sandwiched profile still lies in the classical cone. What is certified is the narrower physical statement that decoherence destroys that cycle, not that the geometry is new.

5 Status

Proved and independently re-verified: Propositions 2.1 and 2.2, Theorems 3.1–3.3, Corollary 3.4, the power-mean identity (6) and its $\lambda = 0$ reversibility, the flatness/permutation-polynomial theorem, the invisibility of the smallest fiber and of symmetry type, superadditivity of C and C_{aff} under Cartesian products — and, for C_{aff} , **strict** superadditivity, witnessed by $x^2 \otimes x \rightarrow x^2 + y$ over $\mathbb{F}_3$ with a gap of at least $1/2$ [affine_rate]. That settles the one form of the synergy question no variational formula could have decided: one affine processor draws a degree-2 part from x^2 and a linear part from x , which neither factor supplies alone.

The first exact affine rates are in hand — $1/2$ for $x^2 \rightarrow x$, $x^2 \rightarrow xy$, $x^2 \rightarrow x^2 + y^2$, and $2/3$ for $x^2 + y^2 \rightarrow x$ — and in the last the Fekete supremum is attained at no $r \leq 2$, so no single-shot computation determines C_{aff} . On the five non-constant quadratic classes over $\mathbb{F}_3$ the tally is now 13 pairs with $C_{\text{aff}} < C$, 4 with $C_{\text{aff}} > C$ (all one-step implementable, where $C_{\text{aff}} = 1$ is forced), 6 equal at 1, and 2 undetermined — so *off the implementable locus* $C_{\text{aff}} \leq C$ holds wherever it is known, a statement equivalent to the open question whether $C_{\text{aff}} = 1$ is possible without $f \preceq_{\text{aff}} g$.

Rationality. Every C_{aff} proved here comes from an N_k that is exactly linear or exactly $\lceil 3k/2 \rceil$, hence eventually exactly linear along arithmetic progressions. Whether that is general is **open**, and is the honest analogue of the rationality of the matrix-multiplication exponent: $C_{\text{aff}} = 1/\lim_k (N_k/k)$ is a limit of rationals with denominator k and need not be rational a priori.

Computed and certified: the cycle censuses and their witnesses; the flat-signature band-cycles; the qubit cycle.

Open and load-bearing. Prove $\Omega \leq 2$, equivalently $|\text{curl } A| \leq (\log 2)/2$; whether C_{aff} cycles on affine orbits; whether any $\lambda < \infty$ is operational; the last two affine brackets over $\mathbb{F}_3$, both with an isotropic resource, now reduced to a single conjecture on the excess of a span of rank- ≤ 2 forms; and whether C_{aff} separates distinct anisotropic square classes over $\mathbb{Q}_p$, where the norm group has index 2 and the finite-field argument has no analogue.

Withdrawn. Three claims of earlier drafts. That the exchange metrics do not fill MET — C_4 is realisable exactly, and the obstruction rested on a stalled search. That $C_{\text{aff}} \leq C$ — false, with the singular-processor counterexample in (3). And that cycles are a zero-temperature artefact — false, with certified cycles at $\lambda = 1$. Each was refuted by the session sent to verify it.